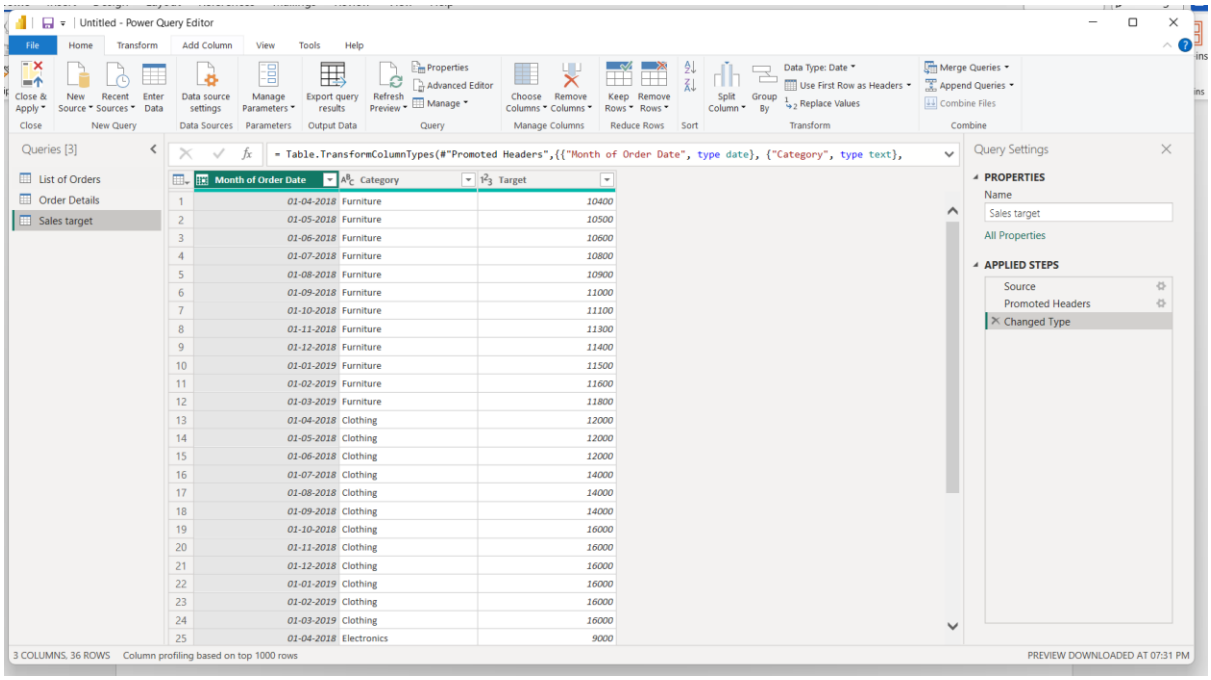
Power BI: Data Transformation and Data Modeling for the E-Commerce Sales Dataset

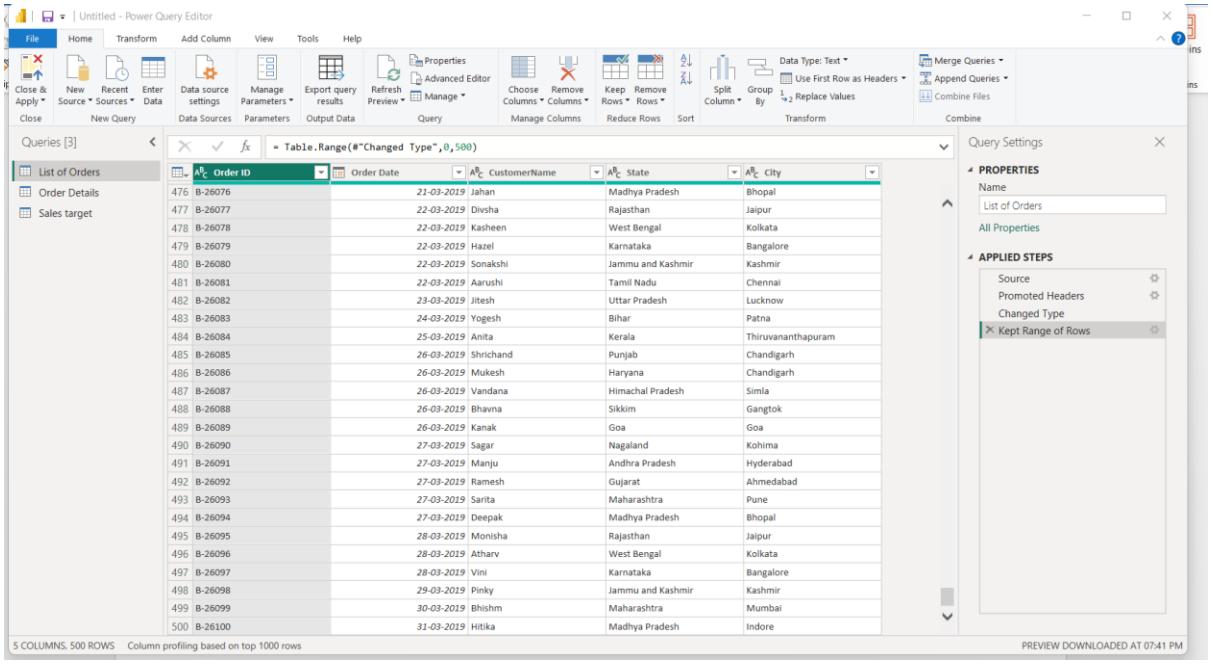
Data Import and Transformation

1. Data Import:

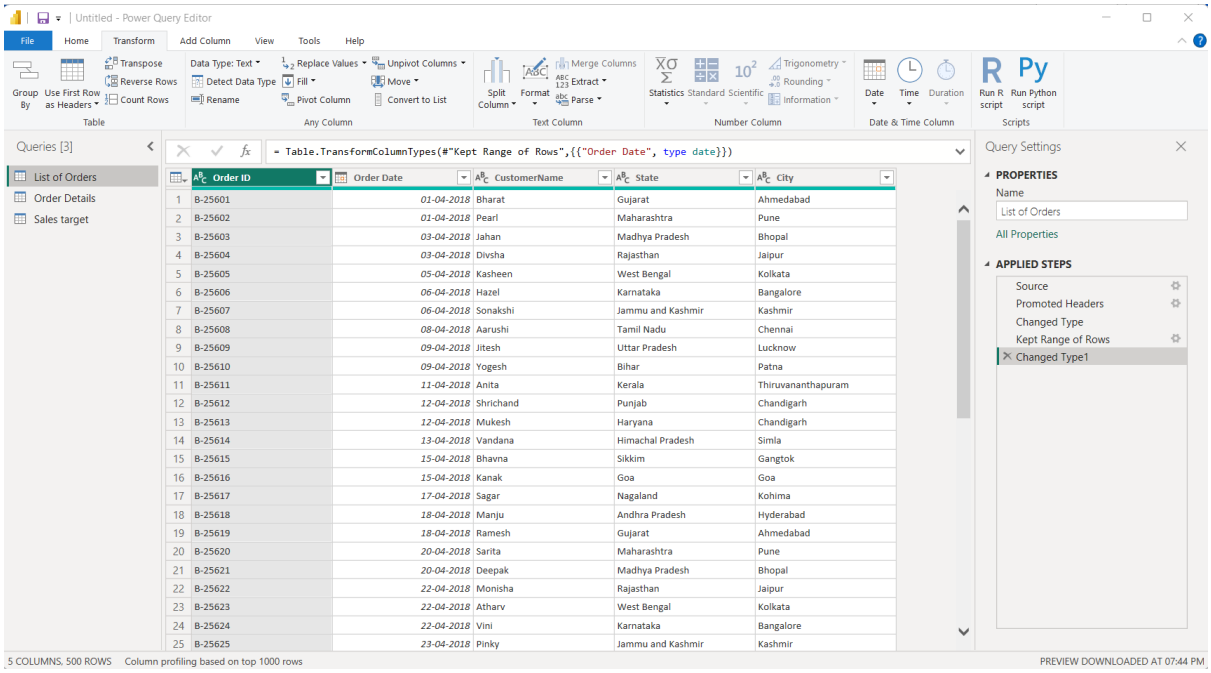


2. Row Limiting & Data Types:

Restrict the List of Orders table to the first 500 rows

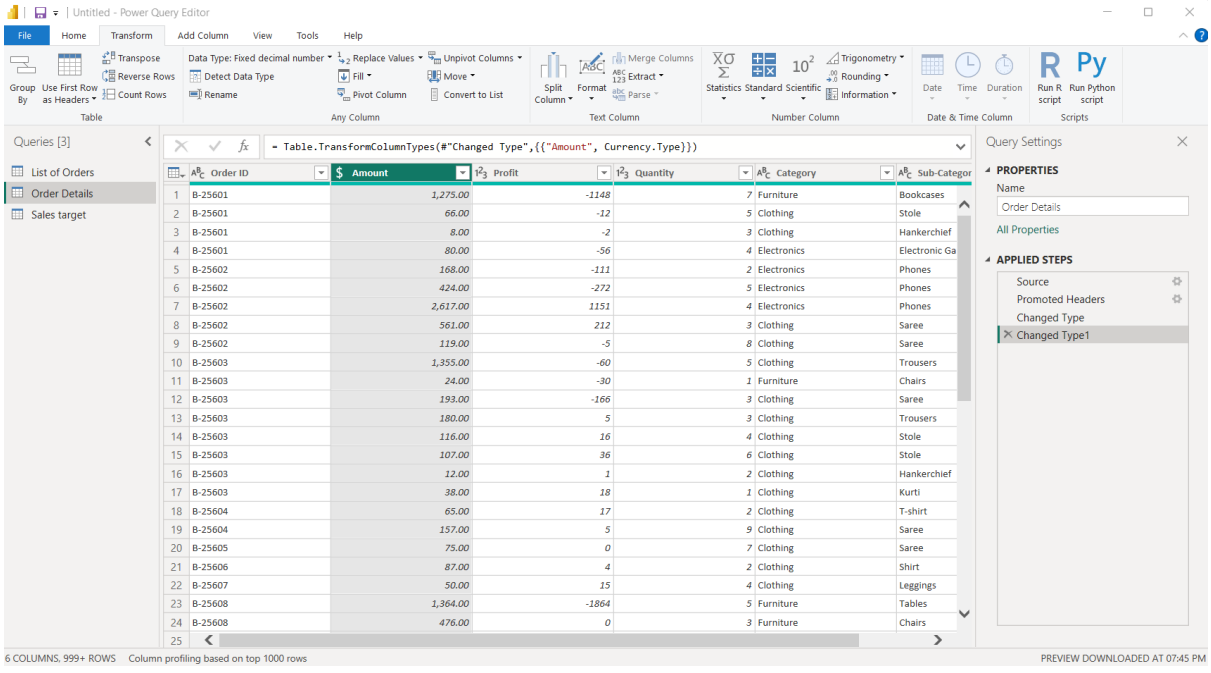


Set the Order Date column to the Date data type

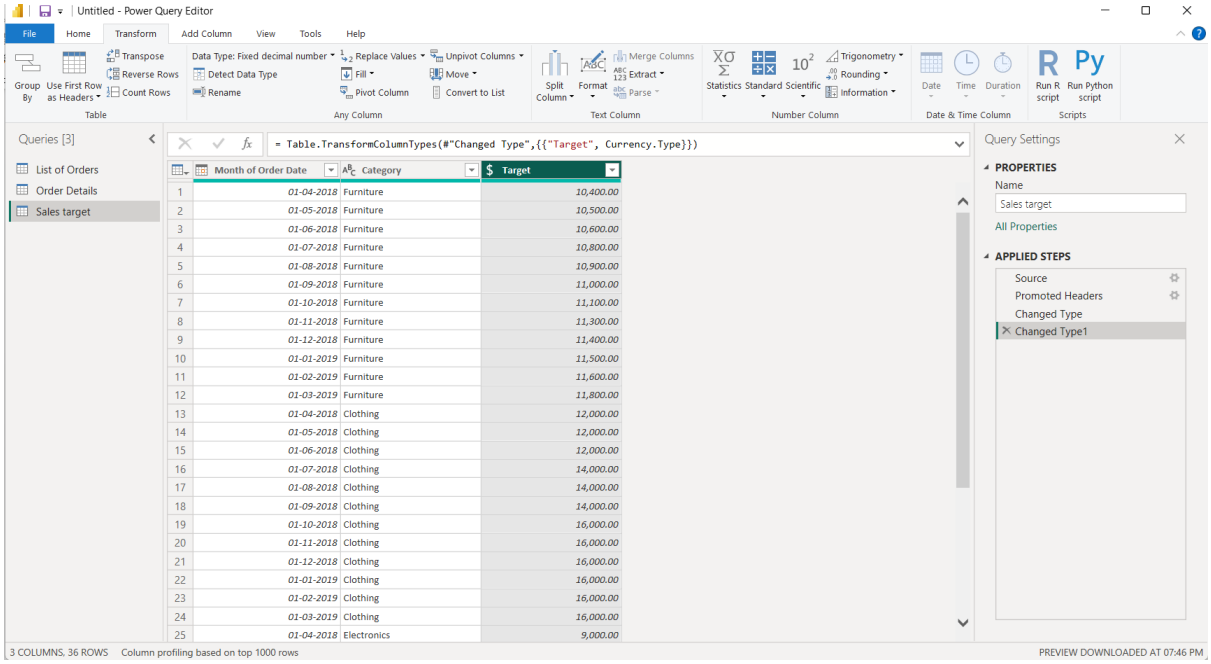


	Order ID	Order Date	CustomerName	State	City
1	B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad
2	B-25602	01-04-2018	Pearl	Maharashtra	Pune
3	B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal
4	B-25604	03-04-2018	Divsha	Rajasthan	Jaipur
5	B-25605	05-04-2018	Kasheen	West Bengal	Kolkata
6	B-25606	06-04-2018	Hazel	Karnataka	Bangalore
7	B-25607	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir
8	B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai
9	B-25609	09-04-2018	Jitesh	Uttar Pradesh	Lucknow
10	B-25610	09-04-2018	Yogesh	Bihar	Patna
11	B-25611	11-04-2018	Anita	Kerala	Thiruvananthapuram
12	B-25612	12-04-2018	Shrichand	Punjab	Chandigarh
13	B-25613	12-04-2018	Mukesh	Haryana	Chandigarh
14	B-25614	13-04-2018	Vandana	Himachal Pradesh	Simla
15	B-25615	15-04-2018	Bhavna	Sikkim	Gangtok
16	B-25616	15-04-2018	Kanak	Goa	Goa
17	B-25617	17-04-2018	Sagar	Nagaland	Kohima
18	B-25618	18-04-2018	Manju	Andhra Pradesh	Hyderabad
19	B-25619	18-04-2018	Ramesh	Gujarat	Ahmedabad
20	B-25620	20-04-2018	Sarita	Maharashtra	Pune
21	B-25621	20-04-2018	Deepak	Madhya Pradesh	Bhopal
22	B-25622	22-04-2018	Monisha	Rajasthan	Jaipur
23	B-25623	22-04-2018	Atharv	West Bengal	Kolkata
24	B-25624	22-04-2018	Vini	Karnataka	Bangalore
25	B-25625	23-04-2018	Pinky	Jammu and Kashmir	Kashmir

Change the Amount and Target columns to fixed decimal numbers



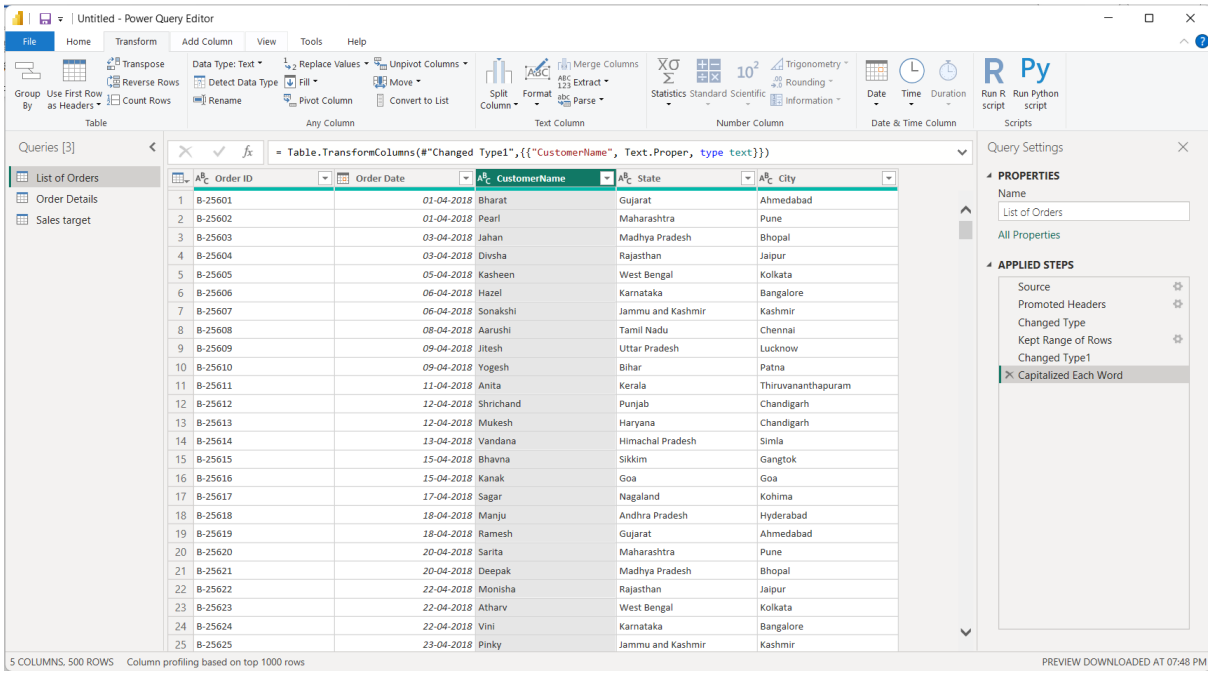
	Order ID	Amount	Profit	Quantity	Category	Sub-Categor
1	B-25601	1,275.00	-1148	7	Furniture	Bookcases
2	B-25601	66.00	-12	5	Clothing	Stole
3	B-25601	8.00	-2	3	Clothing	Hankerchief
4	B-25601	80.00	-56	4	Electronics	Electronic Ga
5	B-25602	168.00	-111	2	Electronics	Phones
6	B-25602	424.00	-272	5	Electronics	Phones
7	B-25602	2,617.00	1151	4	Electronics	Phones
8	B-25602	561.00	212	3	Clothing	Saree
9	B-25602	119.00	-5	8	Clothing	Saree
10	B-25603	1,355.00	-60	5	Clothing	Trousers
11	B-25603	24.00	-30	1	Furniture	Chairs
12	B-25603	193.00	-166	3	Clothing	Saree
13	B-25603	180.00	5	3	Clothing	Trousers
14	B-25603	116.00	16	4	Clothing	Stole
15	B-25603	107.00	36	6	Clothing	Stole
16	B-25603	12.00	1	2	Clothing	Hankerchief
17	B-25603	38.00	18	1	Clothing	Kurti
18	B-25604	65.00	17	2	Clothing	T-shirt
19	B-25604	157.00	5	9	Clothing	Saree
20	B-25605	75.00	0	7	Clothing	Saree
21	B-25606	87.00	4	2	Clothing	Shirt
22	B-25607	50.00	15	4	Clothing	Leggings
23	B-25608	1,364.00	-1864	5	Furniture	Tables
24	B-25608	476.00	0	3	Furniture	Chairs
25						



	Month of Order Date	Category	Target
1	01-04-2018	Furniture	10,400.00
2	01-05-2018	Furniture	10,500.00
3	01-06-2018	Furniture	10,600.00
4	01-07-2018	Furniture	10,800.00
5	01-08-2018	Furniture	10,900.00
6	01-09-2018	Furniture	11,000.00
7	01-10-2018	Furniture	11,100.00
8	01-11-2018	Furniture	11,300.00
9	01-12-2018	Furniture	11,400.00
10	01-01-2019	Furniture	11,500.00
11	01-02-2019	Furniture	11,600.00
12	01-03-2019	Furniture	11,800.00
13	01-04-2018	Clothing	12,000.00
14	01-05-2018	Clothing	12,000.00
15	01-06-2018	Clothing	12,000.00
16	01-07-2018	Clothing	14,000.00
17	01-08-2018	Clothing	14,000.00
18	01-09-2018	Clothing	14,000.00
19	01-10-2018	Clothing	16,000.00
20	01-11-2018	Clothing	16,000.00
21	01-12-2018	Clothing	16,000.00
22	01-01-2019	Clothing	16,000.00
23	01-02-2019	Clothing	16,000.00
24	01-03-2019	Clothing	16,000.00
25	01-04-2018	Electronics	9,000.00

3. Text Formatting

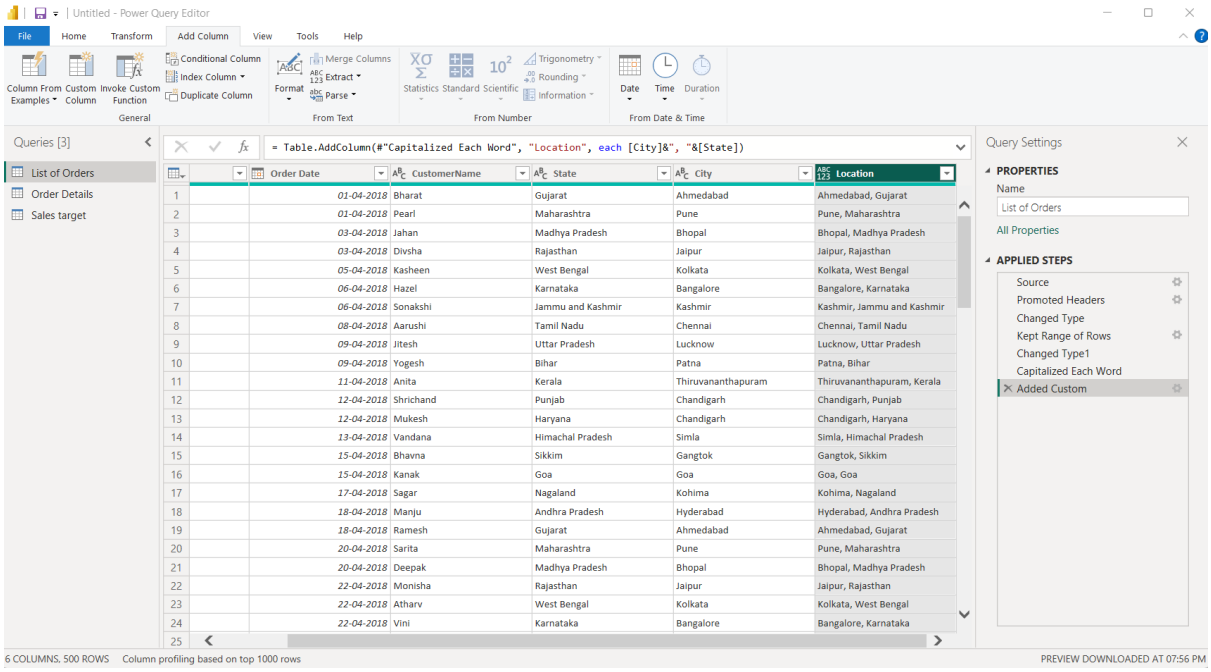
Format the CustomerName column to proper case to ensure consistent capitalization



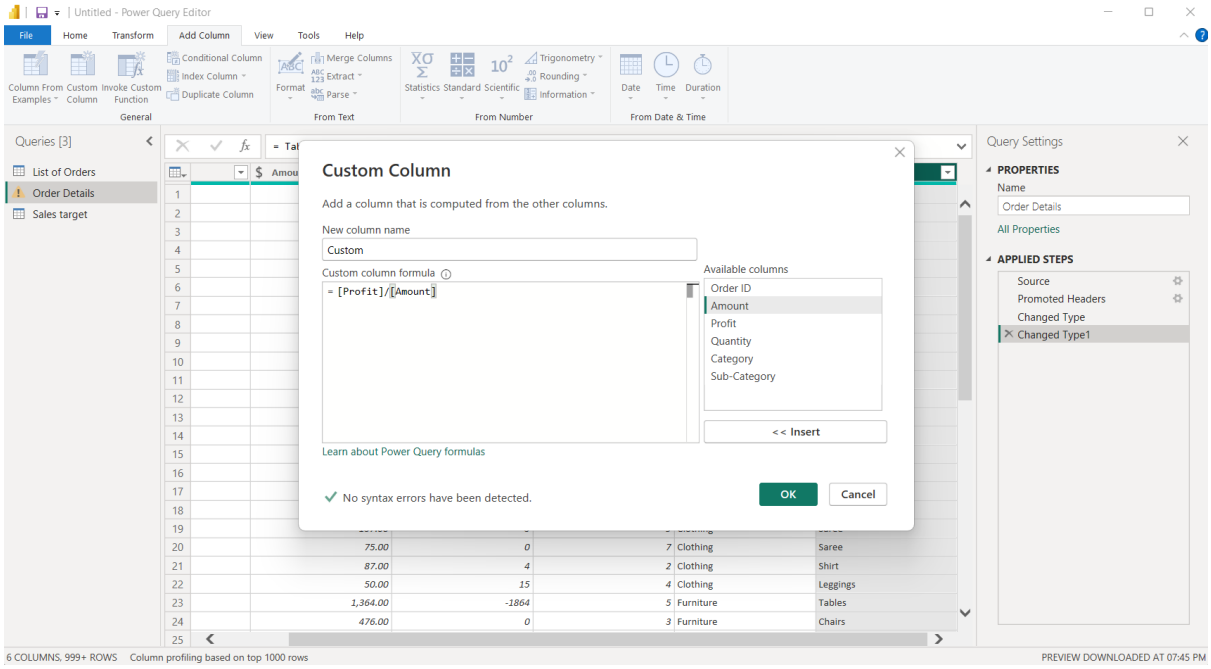
	Order ID	Order Date	CustomerName	State	City
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6	B-25606	06-04-2018	Hazel	Karnataka	Bangalore
7	B-25607	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir
8	B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai
9	B-25609	09-04-2018	Jitesh	Uttar Pradesh	Lucknow
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12	B-25612	12-04-2018	Shrichand	Punjab	Chandigarh
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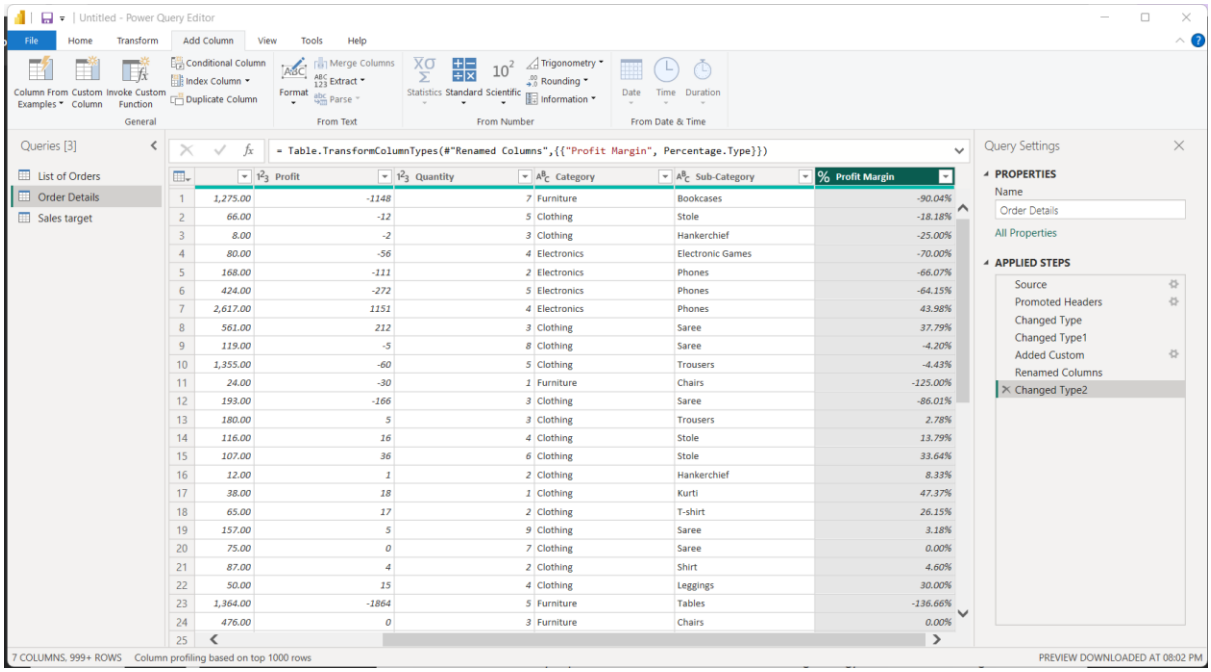
4. Column Creation

Merge the City and State columns to create a new column named Location in the format: City, State



Create a custom column named Profit Margin calculated as: Profit/Amount (expressed as a percentage)

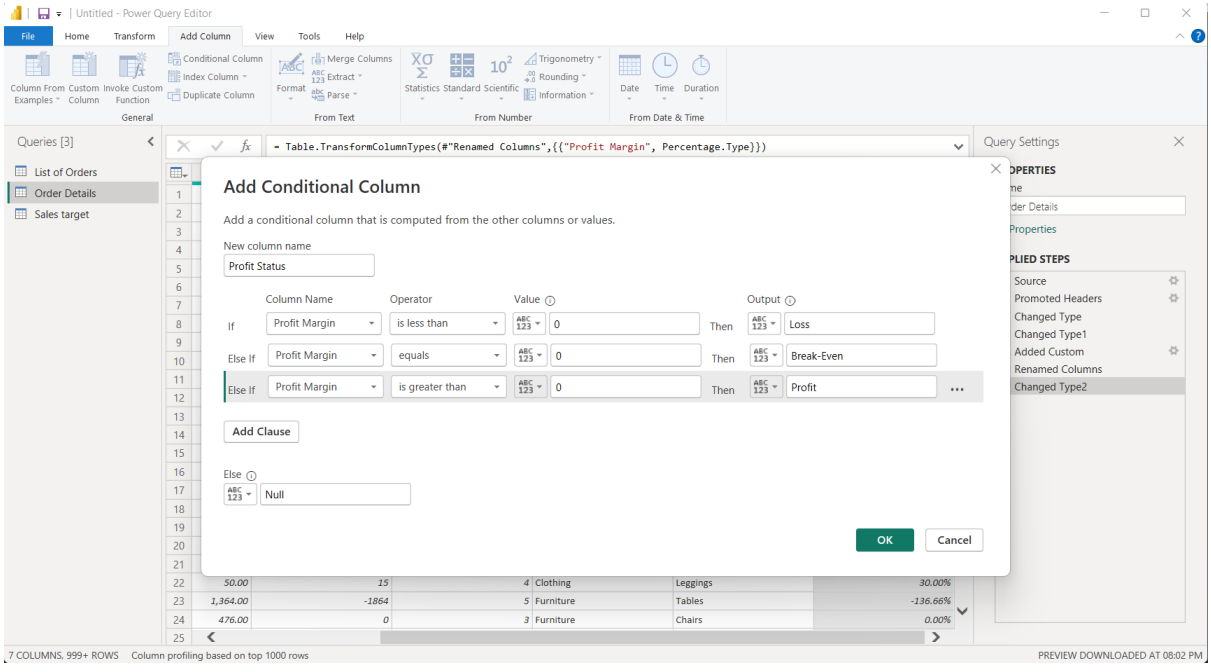


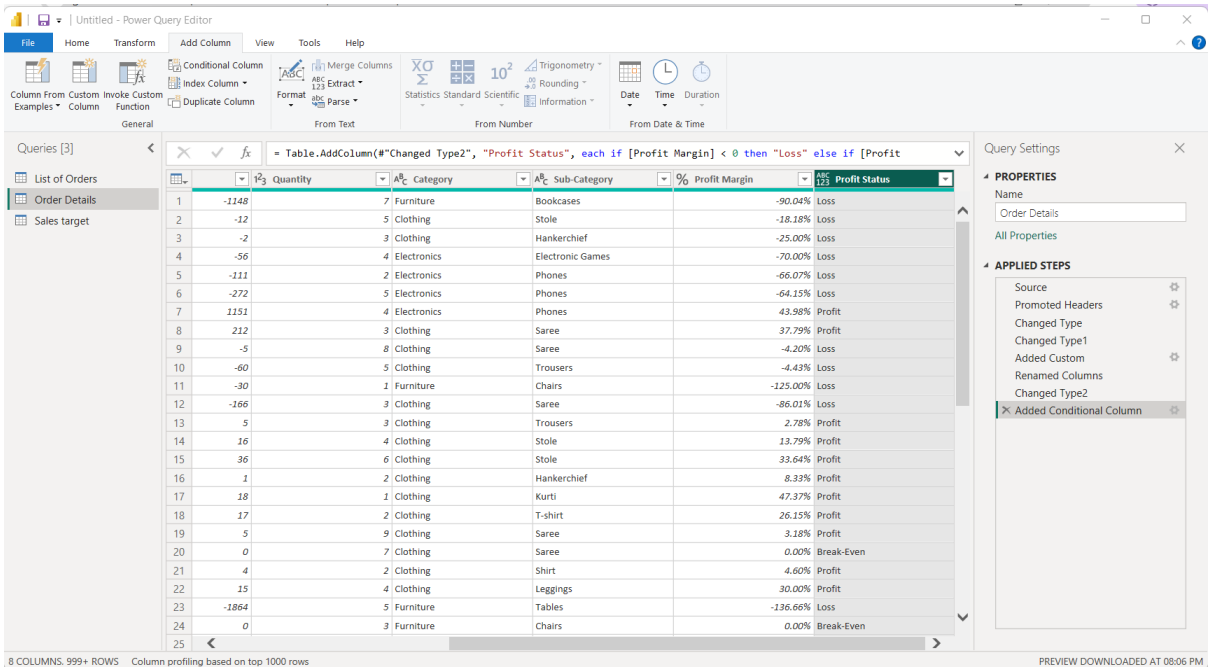


5. Conditional Column

Add a conditional column named Profit Status based on:

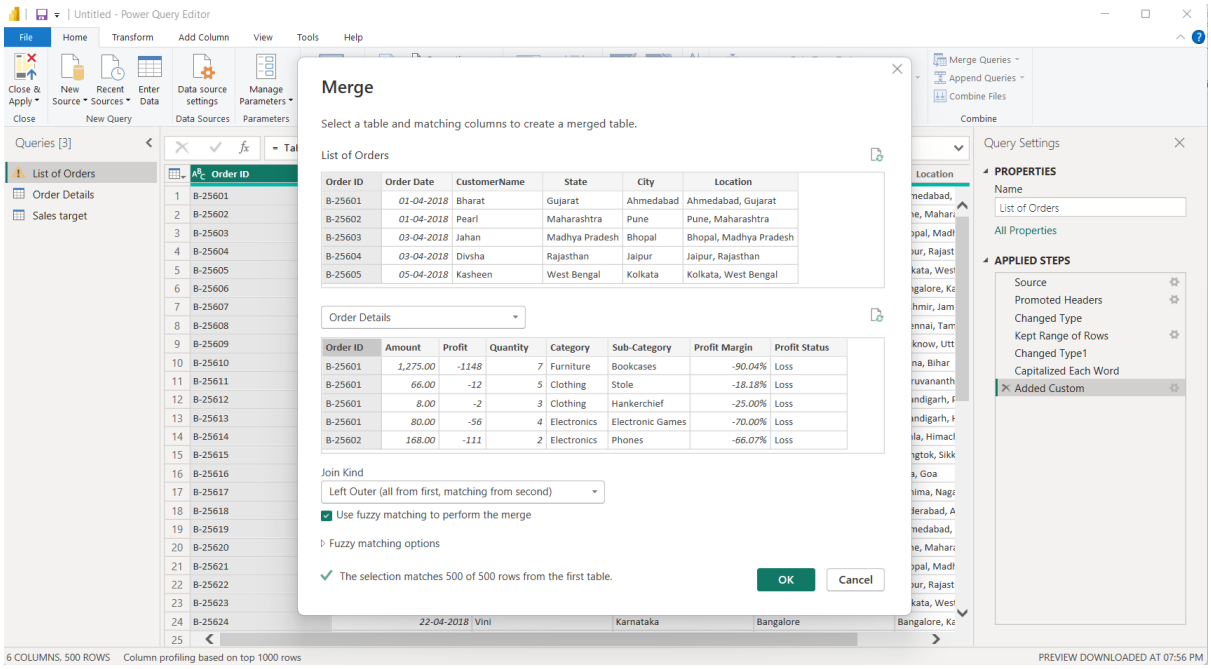
- **Profit < 0 → Loss**
- **Profit = 0 → Break-Even**
- **Profit > 0 → Profit**



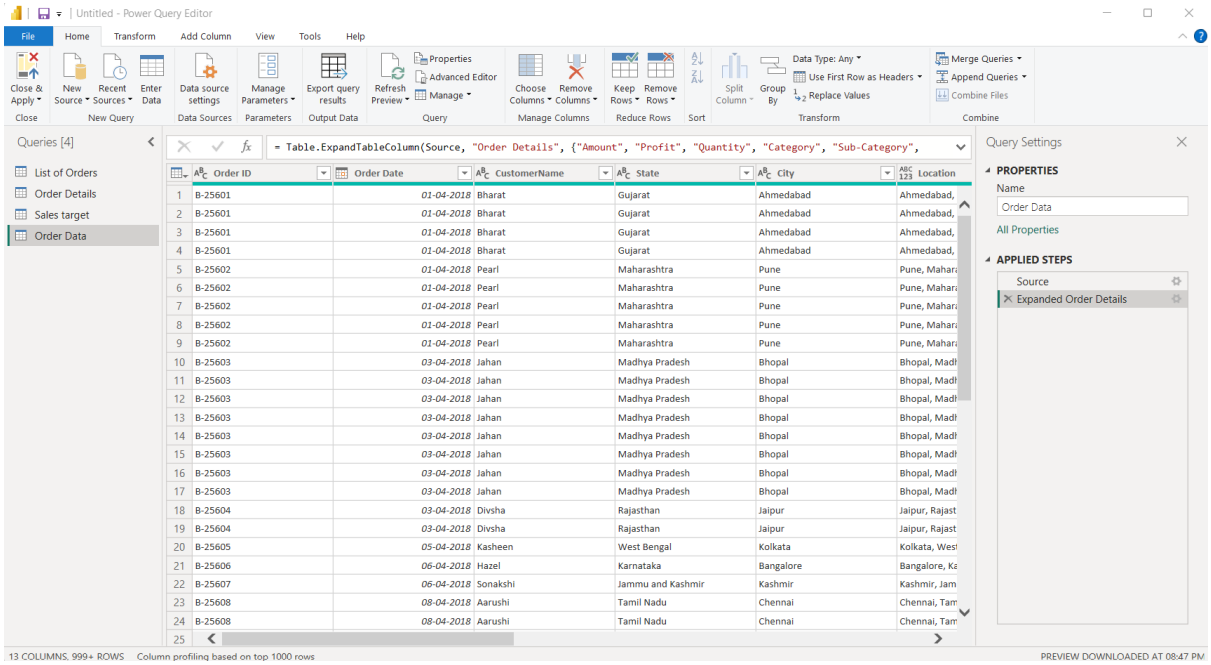


6. Merging Data (Joins)

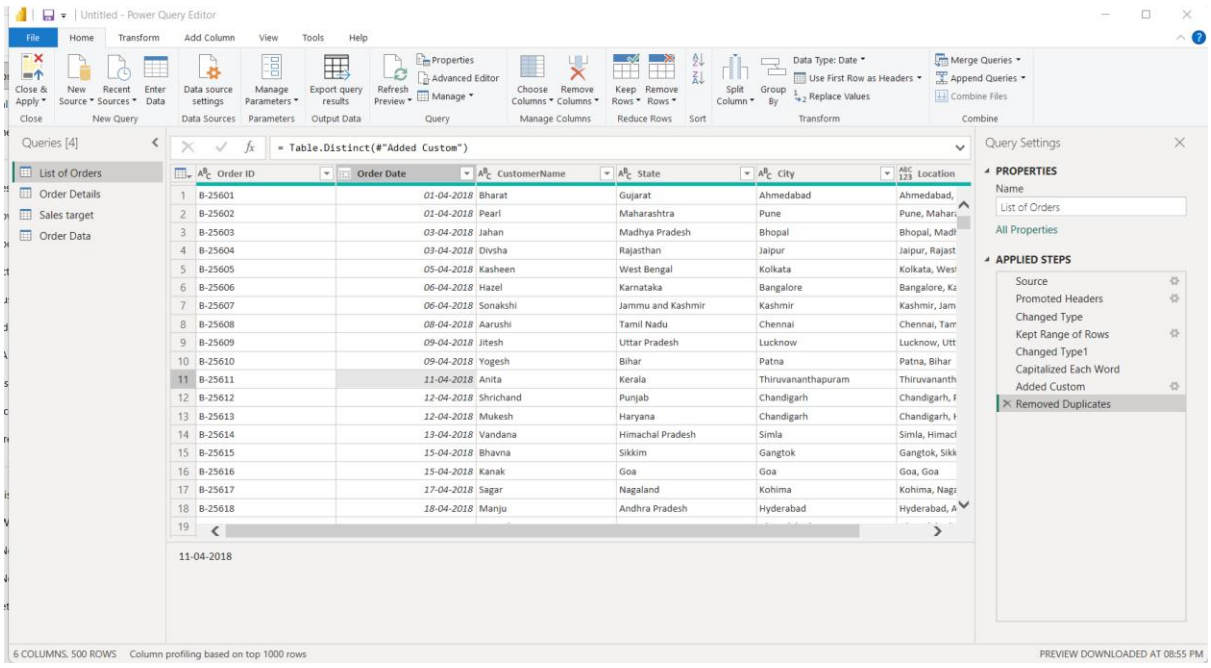
Merge the List of Orders and Order Details tables using Order ID



Name the merged output table as Orders Data

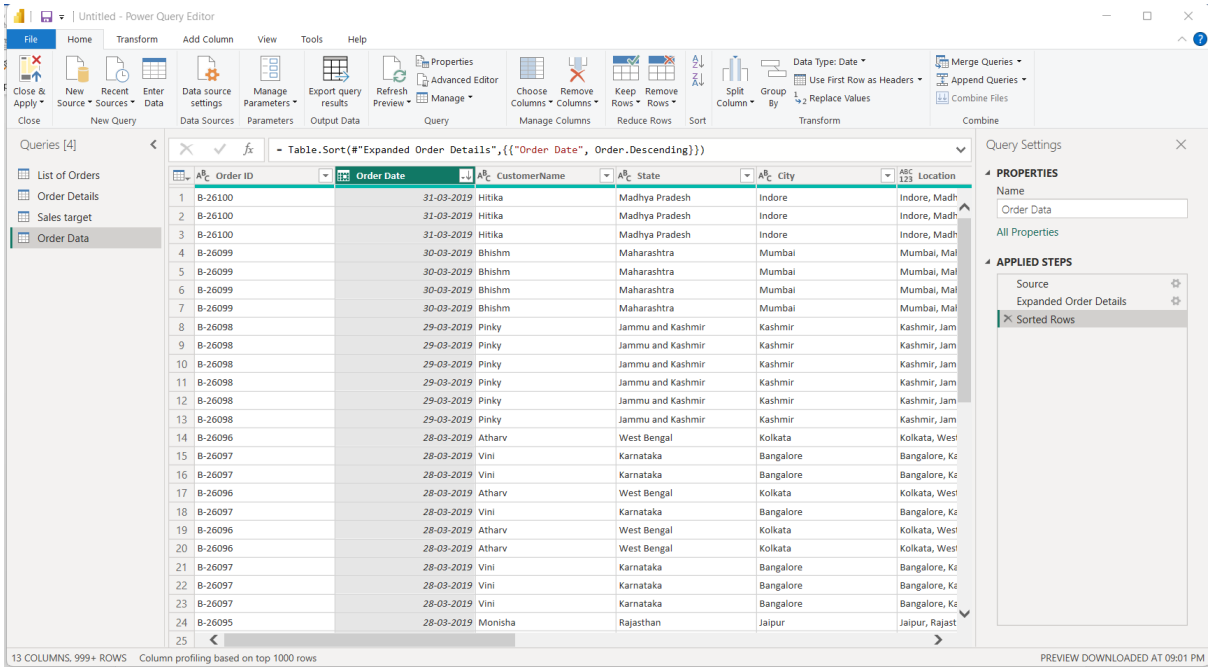


7. Handling Missing Data & Duplicate Data

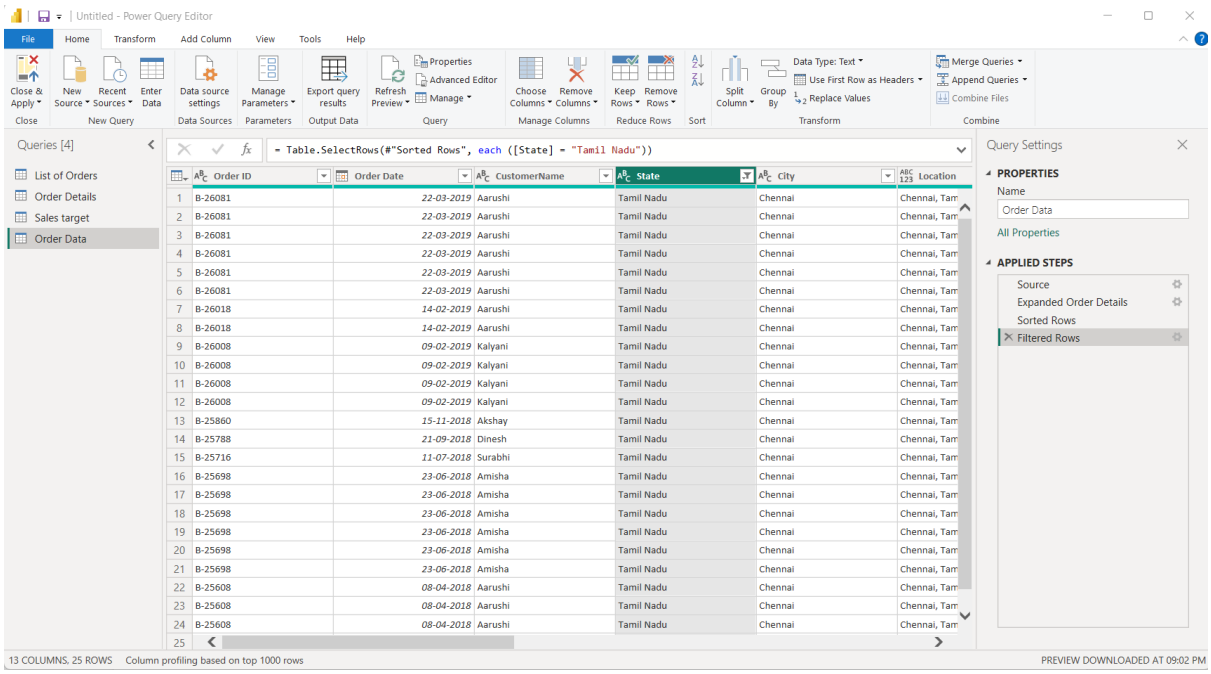


8. Sorting and Filtering Data

Sort records in the Orders Data table by Order Date in descending order



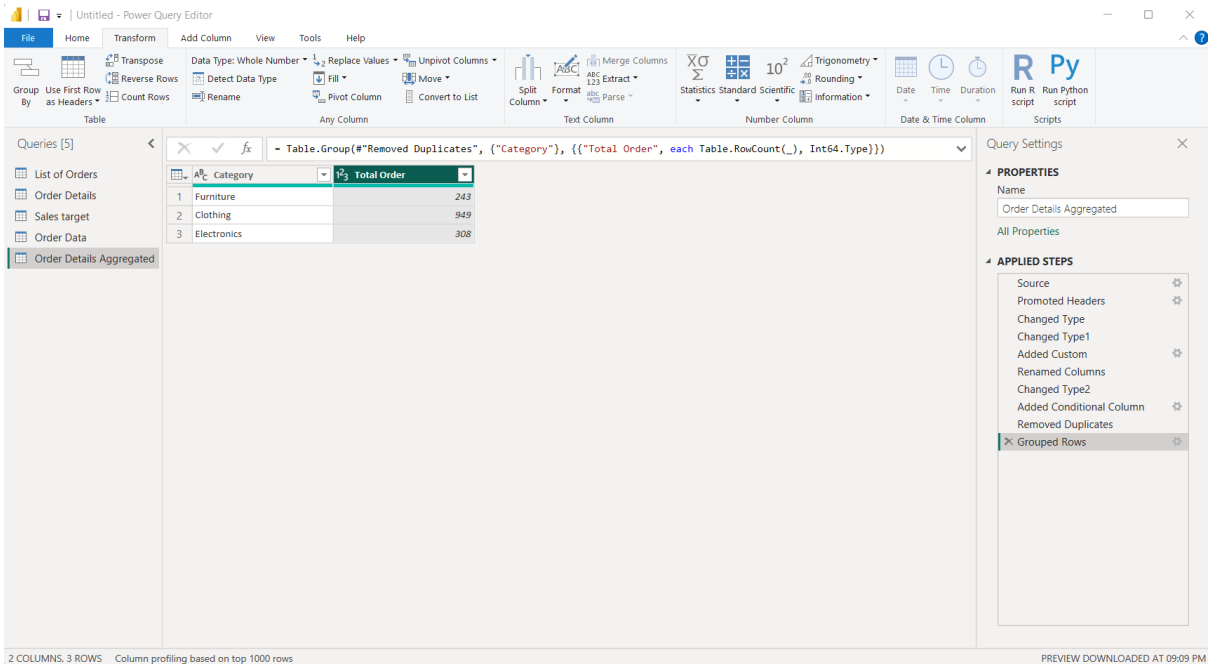
Apply filters to analyze orders from a specific state (e.g., Tamil Nadu)

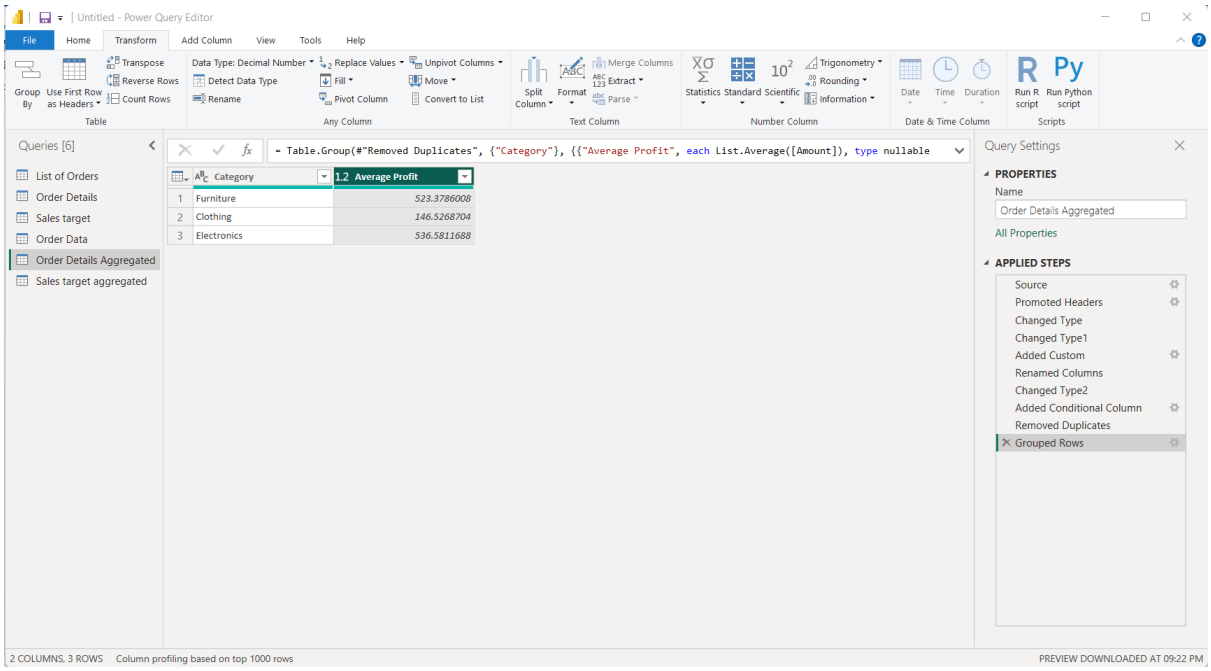


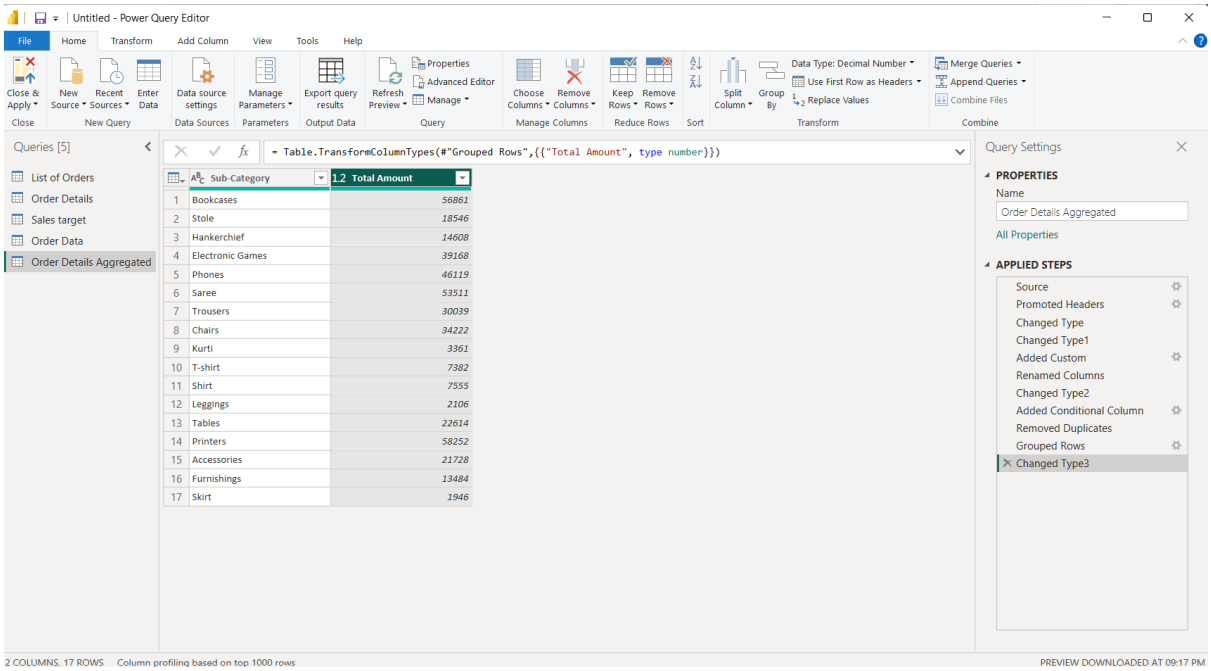
9. Grouping and Aggregating Data

Duplicate the Order Details table and perform the following:

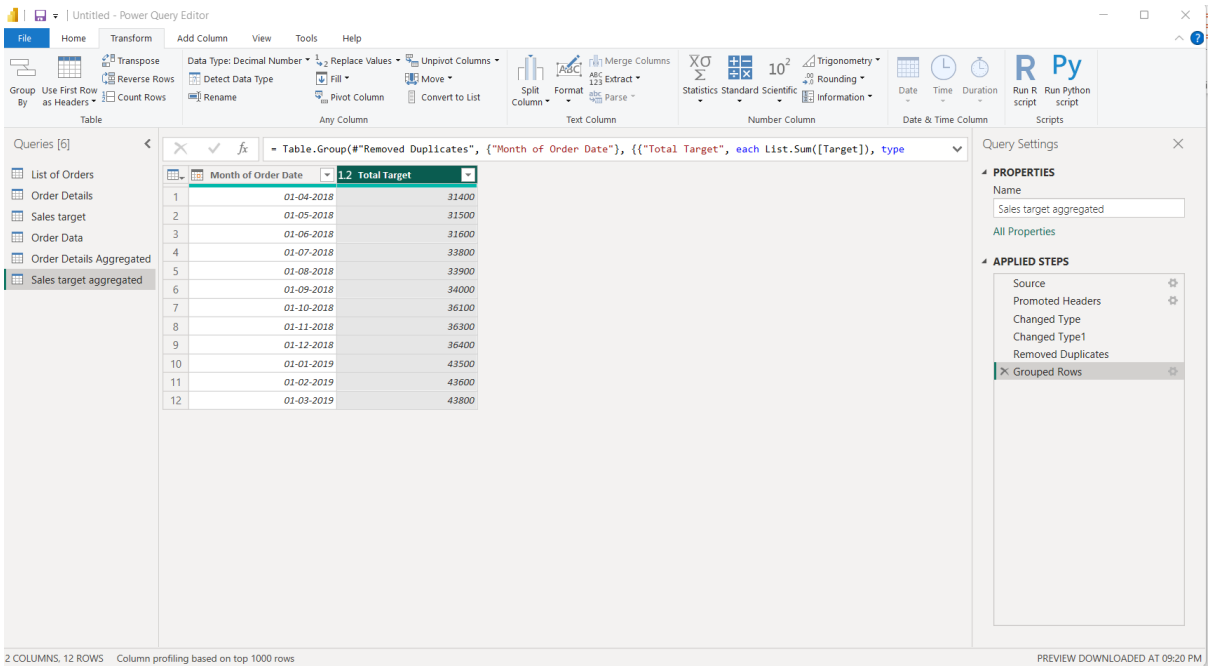
- Count of Order ID
- Average Profit by Category
- Total Amount by Sub-Category


Untitled - Power Query Editor
File
Home
Transform
Add Column
View
Tools
Help
Queries [5]
List of Orders
Order Details
Sales target
Order Data
Order Details Aggregated
= Table.Group(#"Removed Duplicates", {"Category"}, {{"Total Order", each Table.RowCount(_), Int64.Type}})
Category
Total Order
1 Furniture 243
2 Clothing 949
3 Electronics 308
Query Settings
PROPERTIES
Name
Order Details Aggregated
All Properties
APPLIED STEPS
Source
Promoted Headers
Changed Type
Changed Type1
Added Custom
Renamed Columns
Changed Type2
Added Conditional Column
Removed Duplicates
Grouped Rows
2 COLUMNS, 3 ROWS
Column profiling based on top 1000 rows
PREVIEW DOWNLOADED AT 09:09 PM

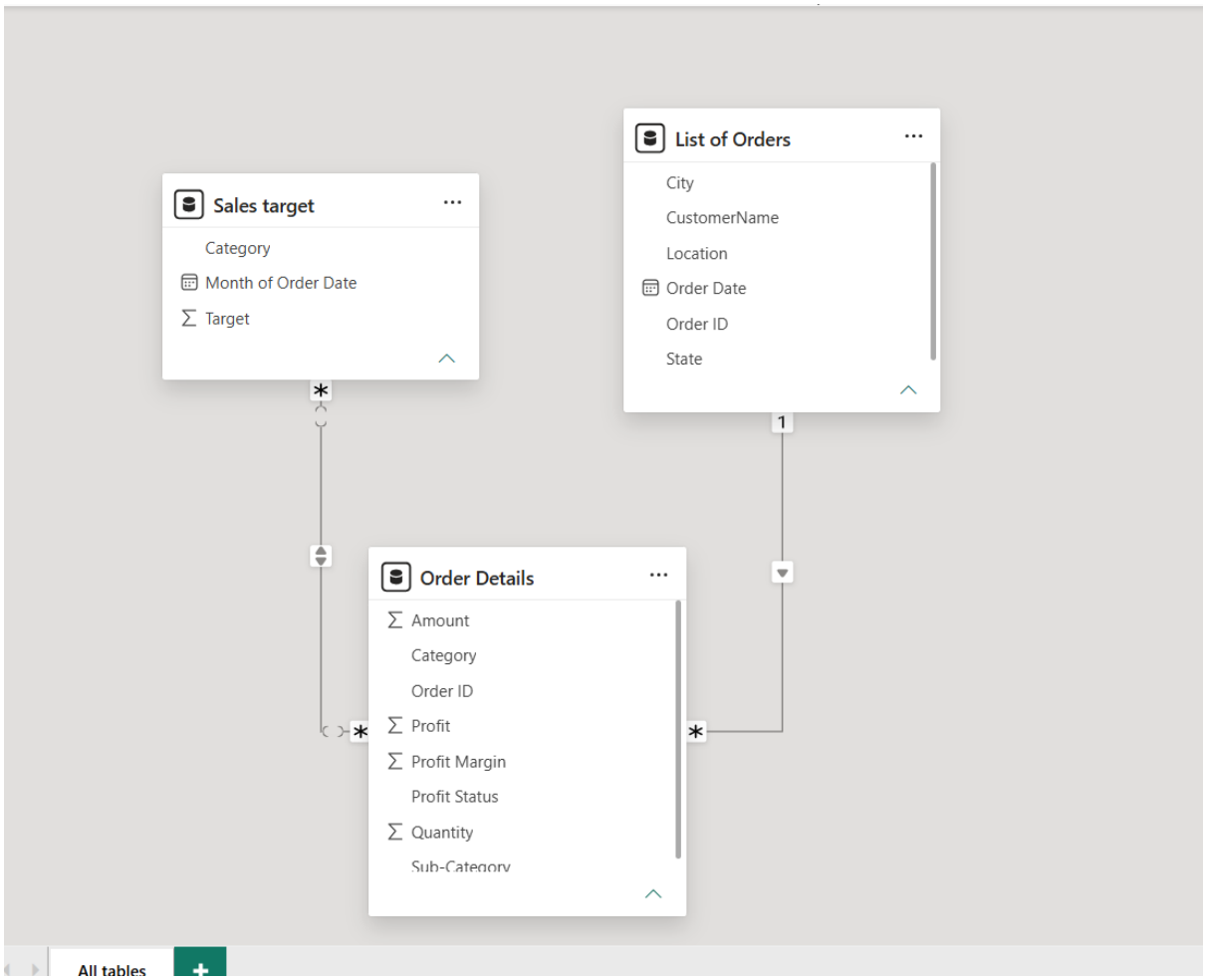

Untitled - Power Query Editor
File
Home
Transform
Add Column
View
Tools
Help
Queries [6]
List of Orders
Order Details
Sales target
Order Data
Order Details Aggregated
Sales target aggregated
= Table.Group(#"Removed Duplicates", {"Category"}, {{"Average Profit", each List.Average([Amount]), type nullable
Category
Average Profit
1 Furniture 523.3786008
2 Clothing 146.5268704
3 Electronics 536.5811688
Query Settings
PROPERTIES
Name
Order Details Aggregated
All Properties
APPLIED STEPS
Source
Promoted Headers
Changed Type
Changed Type1
Added Custom
Renamed Columns
Changed Type2
Added Conditional Column
Removed Duplicates
Grouped Rows
2 COLUMNS, 3 ROWS
Column profiling based on top 1000 rows
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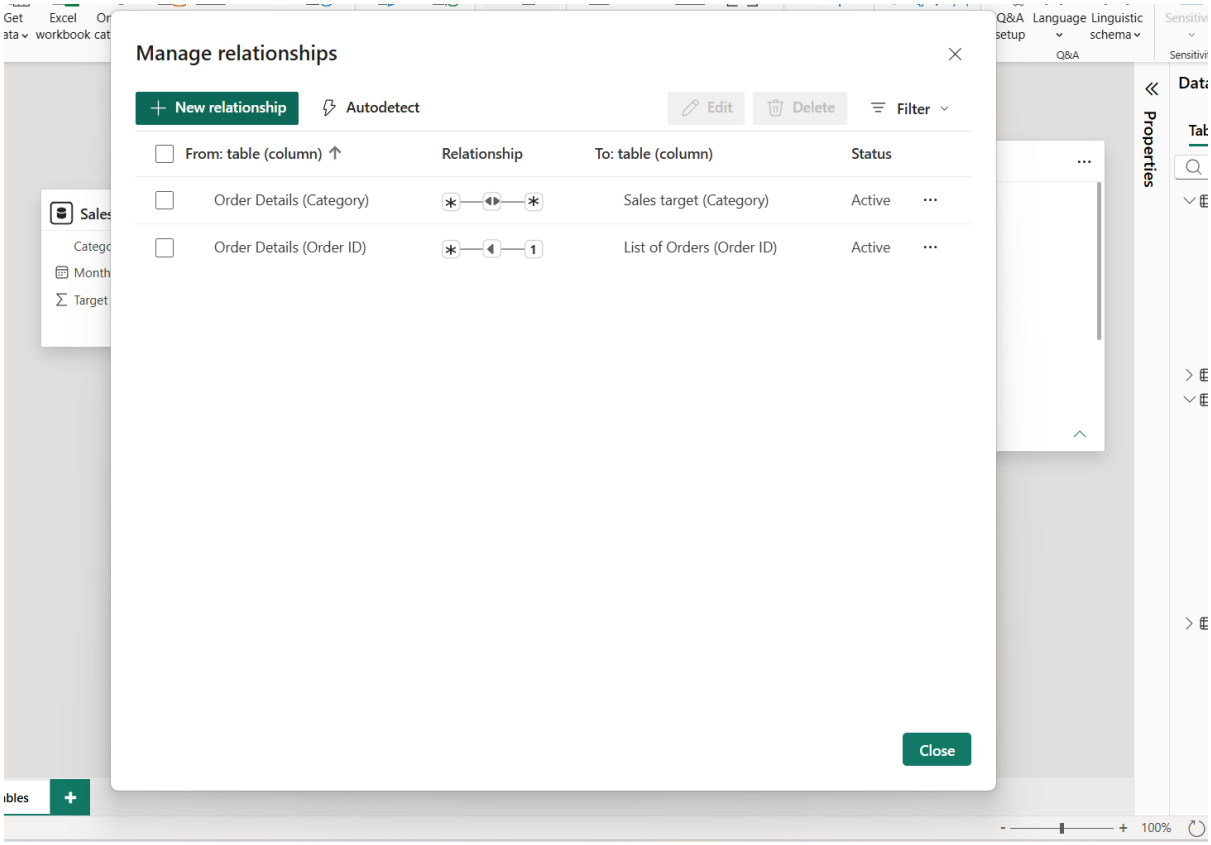


Duplicate the Sales Target table and aggregate the total Target by Month of order date



10. Data Modeling:




Manage relationships
New relationship
Autodetect
Edit
Delete
Filter
From: table (column)
Relationship
To: table (column)
Status
Order Details (Category)
Sales target (Category)
Active
Order Details (Order ID)
List of Orders (Order ID)
Active
Close
Properties
100%